



BITS Pilani
Pilani Campus

Microwave Engineering (ECE F331)

Prof. Pankaj Arora
Department of Electrical and Electronics Engineering



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Pilani Campus

Electromagnetic Fields and Microwave Engineering (ECE F314)

Prof. Pankaj Arora
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Course Handout



Course Description:

Course No.

Course Title

Instructor-in-charge

: ECE F314 or ECE F331

**: Electromagnetic Fields and
Microwave Engineering**

: Prof. Pankaj Arora

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(Chamber: 2210-C, EEE Dept.)

: Prof. Dinesh Rano

Instructors (For Tutorials) :

sec-1: Prof. Dinesh Rano : Room 6102

sec-2: Prof. Pankaj Arora : Room 1202

Text Books:

- 1) TB1: David Pozar, “Microwave Engineering”, 4th edition, John Wiley & Sons, 2012.
- 2) TB2: John D. Kraus and Daniel A. Fleisch, “Electromagnetics”, 5th ed., McGraw-Hill, New York, 1999.

Reference Books:

- i) RB1: David K. Cheng, “Field and Wave Electromagnetics” 2nd ed. Pearson Education, New Delhi, 2009.
- ii) RB2: Samuel Y. Liao, “Microwave devices and circuits,” 3rd ed. , PHI 2008.
- iii) RB3: J. D. Krauss et al., “Antennas and Wave Propagation”, 4th edition, TMH 2010.

Evaluation Scheme (ECE F314):

Component	Duration	Marks (200)	Weightage	Evaluation type
Mid-term Test	90 min	60	30%	Closed Book
Quizzes	TBA	60	30%	Closed Book
Comprehensive Examination	3 hours	80	40%	OB + CB

Evaluation Scheme (ECE F331):

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Mid-term Test	90 min	60	30%	Closed Book
Quizzes	TBA	30	15%	Closed Book
Lab	TBA	30	15%	Close/Open book
Comprehensive Examination	3 hours	80	40%	OB + CB

Why this course?



- Electrical Science, Microelectronic circuits, ADVD etc. (Amplifiers/Oscillators): Low frequency signal
- What about high frequency signal? Consequences ?

→ High frequency signals travel long distances

- For wireless communication: less height of antennas
- To destroy enemies' drones, radars etc. using high frequency signal
- Heating using high frequency signals e.g. Microwave, Oven
- Fusion heating to provide kinetic energy with high frequency signal
- Cargos scanners with high frequency signals
- Medical technologies: for cancer tumors
- Crowd Controls with high frequency signals

What we are going to learn in this course?



- How to generate high frequency signals? {Oscillators??}
- Or
- How to generate high frequency-high power signals?
- How to transmit high frequency signals? {Copper wires??}
- How to radiate high frequency signals? {Antenna??}

Course Content:



- Transmission Lines and Graphical Analysis
- EM Waves and Waveguides
- Microwave Generation and Amplification
- Antennas

Transmitting the signal from Point A to B (Transmission Lines)

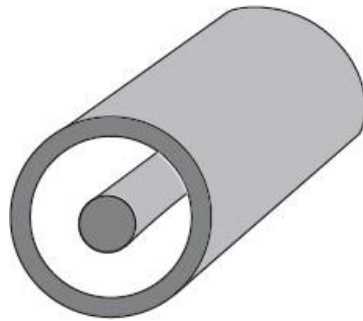
e.g.: Power lines (Copper wires-50 Hz), Telephone Lines(Twisted paired wires-8 to 20kHz), TV Cables (Co-axial wires-MHz), Internet lines (Optical fibers-THz)

Main Difference: Frequency of operation
{Change in Medium}

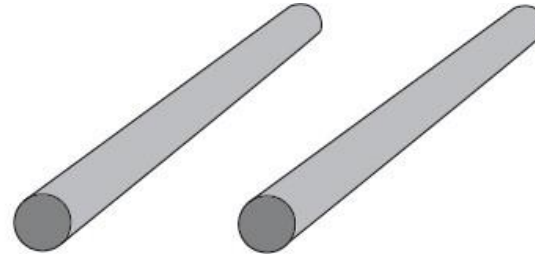
Examples of Transmission Lines:



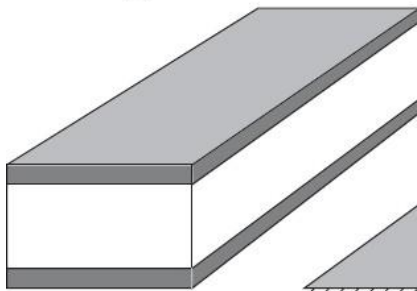
Cross-sectional view



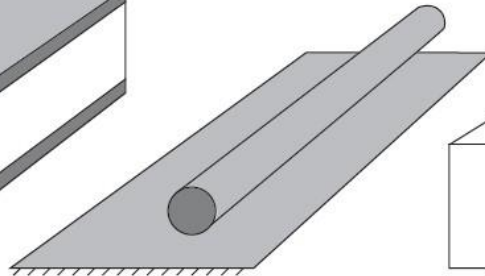
(a)



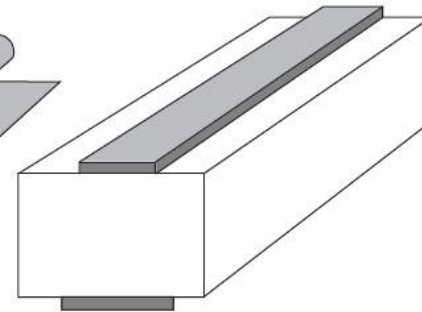
(b)



(c)



(d)



(e)



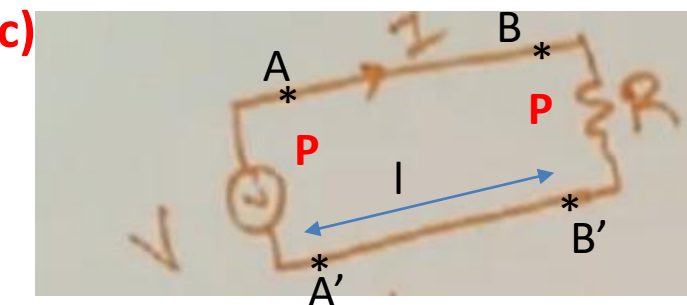
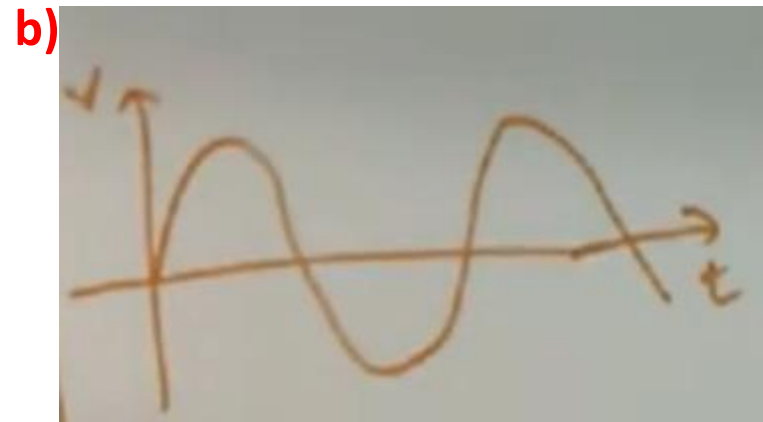
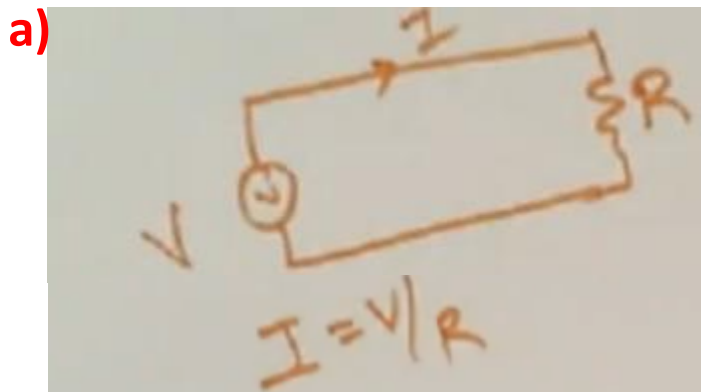
Typical transmission lines in cross-sectional view: (a) coaxial line, (b) two-wire line, (c) planar line, (d) wire above conducting plane, (e) microstrip line.

Example

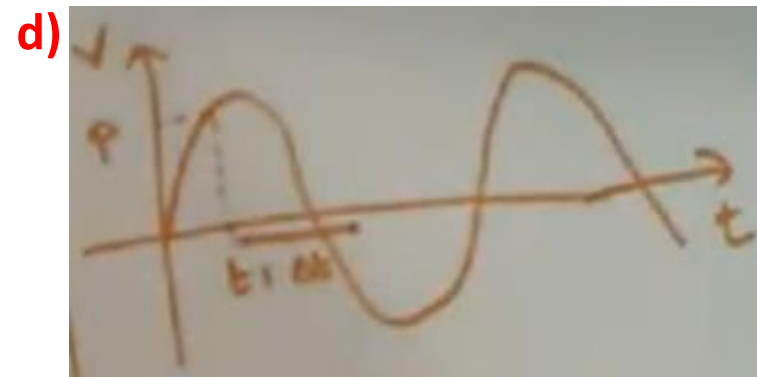


A transmission line can be defined as a device for propagating or guiding energy from one point to another.

Transit time



$$V - \alpha - IR - \alpha = 0$$



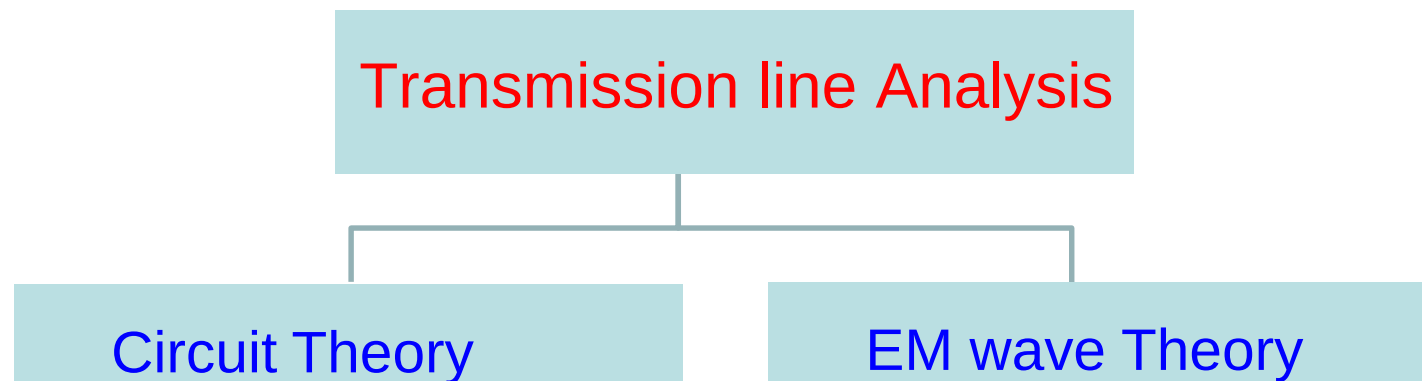
Value of Δt plays important role

High Frequency and longer lines : Circuit theory X

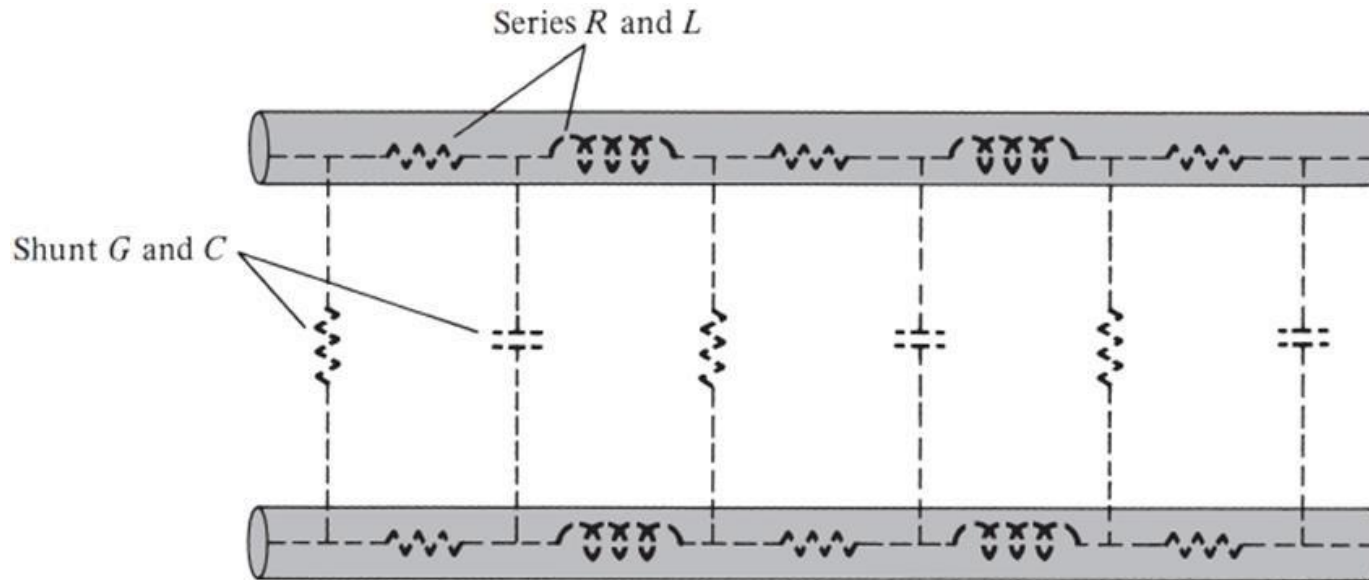
Response of parameters wrt frequency



- **At low frequencies**, the circuit elements are lumped since voltage and current waves affect the entire circuit at the same time.
- **At microwave frequencies**, such treatment of circuit elements is not possible since voltage and current waves do not affect the entire circuit at the same time.



Distributed parameters of a two-conductor transmission line



The impedance parameters L , R , C , and G represent:

L = series inductance per unit length

R = series resistance per unit length

C = shunt capacitance per unit length

G = shunt conductance per unit length.

General Transmission-line Equations



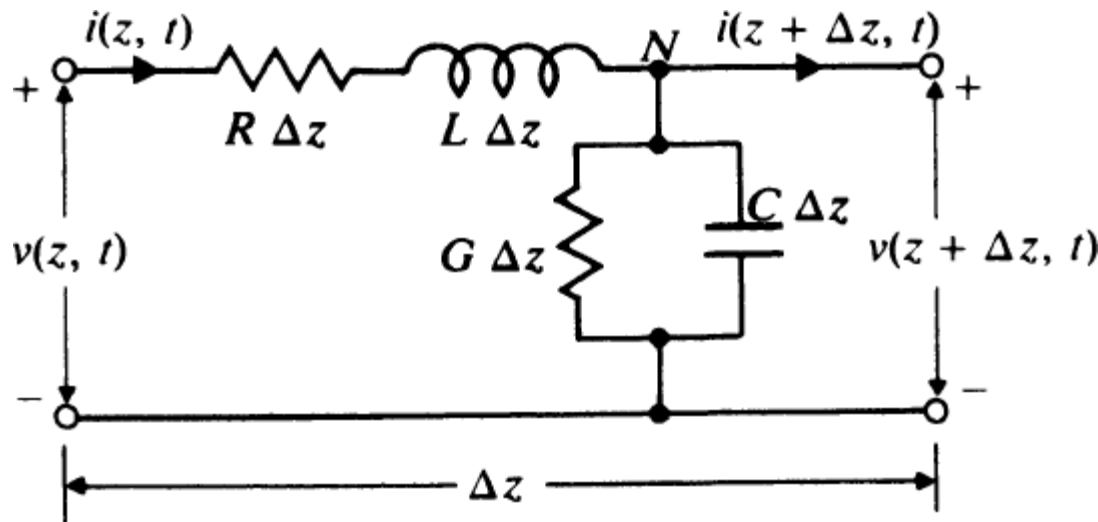
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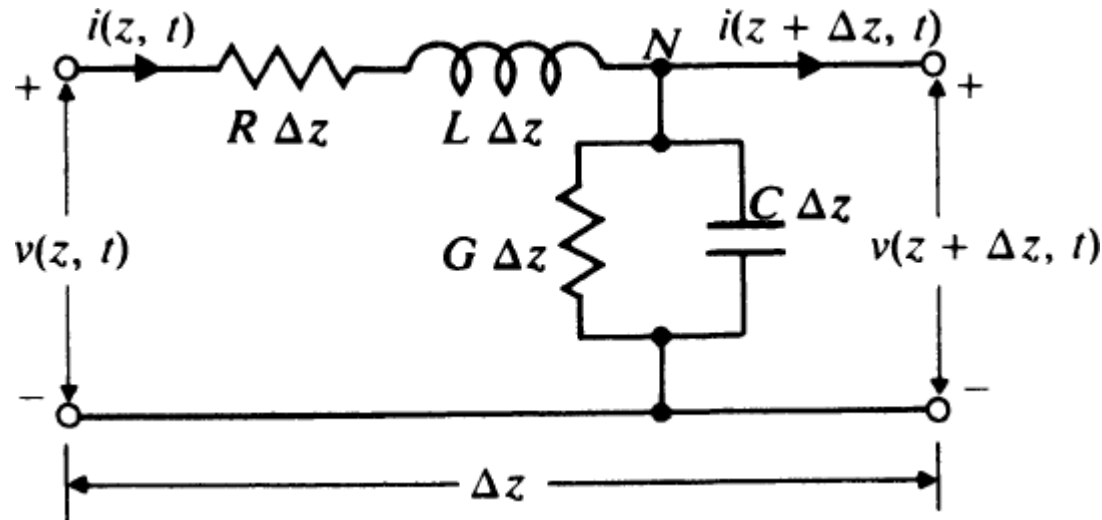
R = series resistance per unit length

C = shunt capacitance per unit length

G = shunt conductance per unit length.



General Transmission-line Equations

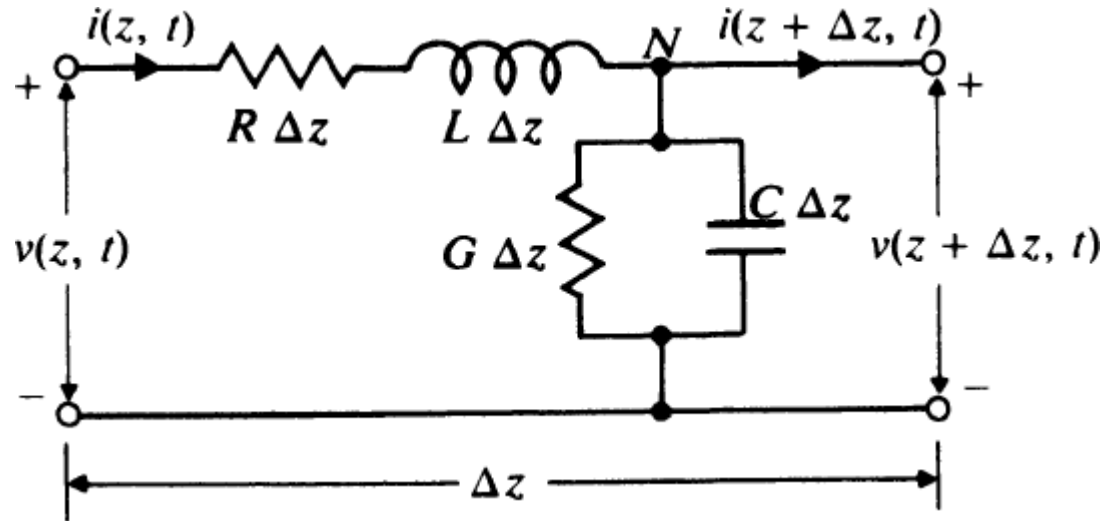


Kirchhoff's voltage law to the outer loop

$$V(z, t) = R \Delta z I(z, t) + L \Delta z \frac{\partial I(z, t)}{\partial t} + V(z + \Delta z, t)$$

$$\frac{V(z + \Delta z, t) - V(z, t)}{\Delta z} = R I(z, t) + L \frac{\partial I(z, t)}{\partial t}$$

General Transmission-line Equations



Kirchoff's current law

$$\begin{aligned}
 I(z, t) &= I(z + \Delta z, t) + \Delta I \\
 &= I(z + \Delta z, t) + G \Delta z V(z + \Delta z, t) + C \Delta z \frac{\partial V(z + \Delta z, t)}{\partial t}
 \end{aligned}$$

or

$$\frac{I(z + \Delta z, t) - I(z, t)}{\Delta z} = G V(z + \Delta z, t) + C \frac{\partial V(z + \Delta z, t)}{\partial t}$$

General Transmission-line Equations



$$-\frac{\partial v(z, t)}{\partial z} = Ri(z, t) + L \frac{\partial i(z, t)}{\partial t}$$

$$-\frac{\partial i(z, t)}{\partial z} = Gv(z, t) + C \frac{\partial v(z, t)}{\partial t}$$

General
Transmission-line
Equations

For Time-Harmonic Dependence:

$$V(z, t) = \text{Re} [V_s(z) e^{j\omega t}]$$

$$I(z, t) = \text{Re} [I_s(z) e^{j\omega t}]$$

$$-\frac{dV_s}{dz} = (R + j\omega L) I_s$$

$$-\frac{dI_s}{dz} = (G + j\omega C) V_s$$

Across the line,
values of V and I
are different

Time-Harmonic
Transmission-line
Equations

General Transmission-line Equations

$$\begin{aligned}-\frac{dV(z)}{dz} &= (R + j\omega L)I(z), \\ -\frac{dI(z)}{dz} &= (G + j\omega C)V(z).\end{aligned}$$

$$\gamma = \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)} \quad (m^{-1})$$

$$\frac{d^2 V(z)}{dz^2} = \gamma^2 V(z)$$

$$\frac{d^2 I(z)}{dz^2} = \gamma^2 I(z),$$

Wave characteristic of an infinite Transmission line



The phasor voltage and current wave equations may be written as

$$\frac{d^2 V(z)}{dz^2} = \gamma^2 V(z)$$

$$\frac{d^2 I(z)}{dz^2} = \gamma^2 I(z),$$

The general solutions to the voltage and current wave equations are

$$\begin{aligned} V(z) &= V^+(z) + V^-(z) \\ &= V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z}, \end{aligned}$$

$$\begin{aligned} I(z) &= I^+(z) + I^-(z) \\ &= I_0^+ e^{-\gamma z} + I_0^- e^{\gamma z}, \end{aligned}$$

+z-directed waves

-z-directed waves

Propagation constant:

$$\gamma = \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)} \quad (m^{-1})$$

Characteristic impedance



$$-\frac{dV(z)}{dz} = (R + j\omega L)I(z),$$

$$-\frac{dI(z)}{dz} = (G + j\omega C)V(z).$$

$$\begin{aligned} V(z) &= V^+(z) + V^-(z) \\ &= V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z}, \end{aligned}$$

$$\begin{aligned} I(z) &= I^+(z) + I^-(z) \\ &= I_0^+ e^{-\gamma z} + I_0^- e^{\gamma z}, \end{aligned}$$

$$\frac{d}{dz} (V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z}) = -(R + j\omega L) (I_0^+ e^{-\gamma z} + I_0^- e^{\gamma z})$$

$$\underbrace{V_0^+ (-\gamma) e^{-\gamma z}} + \underbrace{(\gamma) V_0^- e^{\gamma z}} = -\underbrace{(R + j\omega L) I_0^+ e^{-\gamma z}} - \underbrace{(R + j\omega L) I_0^- e^{\gamma z}}$$

$$\frac{V_0^+}{I_0^+} = \frac{(R + j\omega L)}{\gamma} = -\frac{V_0^-}{I_0^-} \Rightarrow Z_0$$

$$-\underbrace{(R + j\omega L) I_0^- e^{\gamma z}}$$

Wave characteristic of an infinite Transmission line



Characteristic impedance:

The **characteristic impedance** Z_o of the line is the ratio of positively traveling voltage wave to current wave at any point on the line.

$$Z_o = \frac{V_o^+}{I_o^+} = -\frac{V_o^-}{I_o^-} = \frac{R + j\omega L}{\gamma} = \frac{\gamma}{G + j\omega C}$$

$$= \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

Propagation constant:

$$\gamma = \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)} \quad (m^{-1})$$

Secondary Constants:

Propagation constant

Characteristic impedance

Classification of Transmission lines



➤ **Lossless line** ($R = 0$ and $G = 0$)

A **transmission line** is said to be **lossless** if the conductors of the line are perfect ($\sigma_c \approx \infty$) and the dielectric medium separating them is lossless ($\sigma \approx 0$).

$$\alpha = 0, \quad \gamma = j\beta = j\omega \sqrt{LC}$$

$$u = \frac{\omega}{\beta} = \frac{1}{\sqrt{LC}} = f\lambda$$

$$X_o = 0, \quad Z_o = R_o = \sqrt{\frac{L}{C}}$$

Wave characteristic of Transmission line (lossless)



$$V(z) = V^+(z) + V^-(z) \\ = \boxed{V_0^+ e^{-\gamma z}} + V_0^- e^{\gamma z},$$

[For Time harmonic variation]

$$v(z) = V_0^+ e^{-\gamma z} e^{j\omega t} \\ = V_0^+ e^{-\alpha z} e^{-j\beta z} e^{j\omega t}$$

For lossless line $\alpha = 0$

$$V(z) = V_0^+ e^{j(\omega t - \beta z)} \text{ or } V_0^+ e^{-j(\beta z - \omega t)}$$

$$\underline{\underline{\text{Re}[v(z)] = V_0^+ \cos(\beta z - \omega t)}} \quad \text{Wave nature}$$

Wave characteristic of Transmission line (not lossless)



$$V(z) = V^+(z) + V^-(z) \\ = \underbrace{V_0^+ e^{-\gamma z}} + V_0^- e^{\gamma z},$$

$$v(z) = V_0^+ e^{-\alpha^2 - j\beta^2} e^{j\omega t}$$

Time harmonic e

$$v(z) = V_0^+ e^{-\alpha z} e^{-j\beta z - j\omega t}$$

What is $\beta \rightarrow ??$

units

What is $\alpha \rightarrow ??$

units

$$e^{-\alpha^2} \Rightarrow$$

$$[\text{Np/m} = -8.68 \text{ dB}] ??$$

Classification of Transmission lines



➤ Distortionless line ($R/L=G/C$)

A **distortionless line** is one in which the attenuation constant α is frequency independent while the phase constant β is linearly dependent on frequency.

$$\begin{aligned}\gamma &= \sqrt{RG \left(1 + \frac{j\omega L}{R}\right) \left(1 + \frac{j\omega C}{G}\right)} \\ &= \sqrt{RG} \left(1 + \frac{j\omega C}{G}\right) = \alpha + j\beta\end{aligned}$$

$$\alpha = \sqrt{RG}, \quad \beta = \omega \sqrt{LC}$$

$$Z_o = \sqrt{\frac{R(1 + j\omega L/R)}{G(1 + j\omega C/G)}} = \sqrt{\frac{R}{G}} = \sqrt{\frac{L}{C}} = R_o + jX_o$$

$$R_o = \sqrt{\frac{R}{G}} = \sqrt{\frac{L}{C}}, \quad X_o = 0$$

$$u = \frac{\omega}{\beta} = \frac{1}{\sqrt{LC}} = f\lambda$$

Classification of Transmission lines



Low loss line ($R \ll \omega L$, $G \ll \omega C$)

$$\begin{aligned}\gamma &= \alpha + j\beta = j\omega\sqrt{LC}\left(1 + \frac{R}{j\omega L}\right)^{1/2}\left(1 + \frac{G}{j\omega C}\right)^{1/2} \\ &\cong j\omega\sqrt{LC}\left(1 + \frac{R}{2j\omega L}\right)\left(1 + \frac{G}{2j\omega C}\right) \\ &\cong j\omega\sqrt{LC}\left[1 + \frac{1}{2j\omega}\left(\frac{R}{L} + \frac{G}{C}\right)\right];\end{aligned}$$

$$\alpha \cong \frac{1}{2}\left(R\sqrt{\frac{C}{L}} + G\sqrt{\frac{L}{C}}\right)$$

$$\beta \cong \omega\sqrt{LC} \quad (\text{approximately a linear function of } \omega).$$

Classification of Transmission lines



Low loss line ($R \ll \omega L$, $G \ll \omega C$)

$$\alpha_p = \frac{\omega}{\beta} \cong \frac{1}{\sqrt{LC}}$$

$$Z_0 = R_0 + jX_0 = \sqrt{\frac{L}{C}} \left(1 + \frac{R}{j\omega L}\right)^{1/2} \left(1 + \frac{G}{j\omega C}\right)^{-1/2}$$

$$\cong \sqrt{\frac{L}{C}} \left[1 + \frac{1}{2j\omega} \left(\frac{R}{L} - \frac{G}{C}\right)\right];$$

$$R_0 \cong \sqrt{\frac{L}{C}}$$

$$X_0 \cong -\sqrt{\frac{L}{C}} \frac{1}{2\omega} \left(\frac{R}{L} - \frac{G}{C}\right)$$

Reflection coefficient



The general solutions to the voltage and current wave equations are

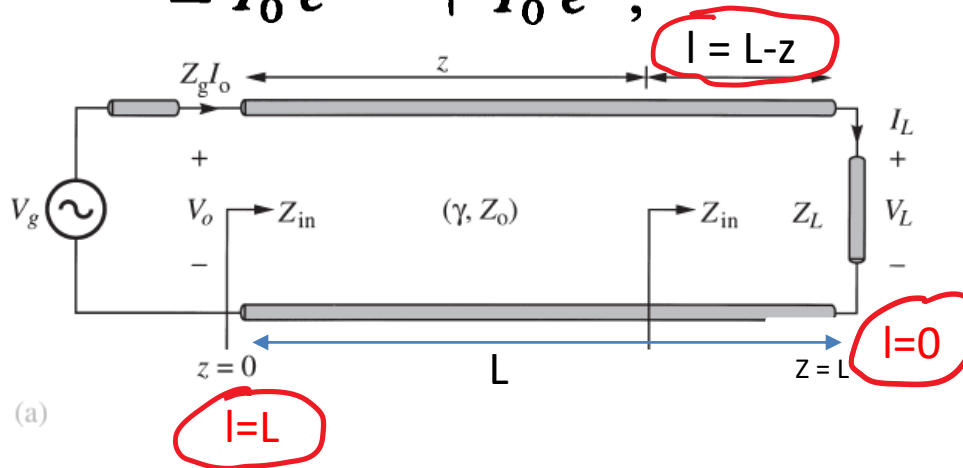
$$V(z) = V^+(z) + V^-(z)$$

$$I(z) = I^+(z) + I^-(z) \quad ;$$

$$= I_0^+ e^{-\gamma z} + I_0^- e^{\gamma z},$$

$$V_s(z) = V_o^+ e^{-\gamma z} + V_o^- e^{\gamma z}$$

$$I_s(z) = \frac{V_o^+}{Z_o} e^{-\gamma z} - \frac{V_o^-}{Z_o} e^{\gamma z}$$



$$Z_{in} = \frac{V_s(z)}{I_s(z)} = Z_o \frac{V_o^+ e^{-\gamma z} + V_o^- e^{\gamma z}}{V_o^+ e^{-\gamma z} - V_o^- e^{\gamma z}}$$

**Voltage reflection coefficient
(at load, l)**

$$\Gamma(l) = \frac{V_o^- e^{-\gamma l}}{V_o^+ e^{\gamma l}} \approx \frac{V_o^-}{V_o^+} e^{-2\gamma l} = \text{Reflection coefficient}$$

$$\left[\begin{aligned} V_s(l) &= V_o^+ e^{-\gamma l} + V_o^- e^{\gamma l} \\ I_s(l) &= \frac{V_o^+}{Z_o} e^{-\gamma l} - \frac{V_o^-}{Z_o} e^{\gamma l} \end{aligned} \right]$$

Reflection coefficient



$$\Gamma(l) = \frac{V_o^- e^{-\gamma l}}{V_o^+ e^{\gamma l}} \approx \frac{V_o^-}{V_o^+} e^{-2\gamma l} = \text{Reflection coefficient}$$

at $l=0$ $\left[\Gamma_{l=0} = \frac{V_o^-}{V_o^+} \right]$

So $\left[\Gamma(l) = (\Gamma_{l=0}) e^{-2\gamma l} \right] \Rightarrow \underline{\text{Important}}$

$$\left[\begin{aligned} V_s(l) &= V_o^+ e^{+\gamma l} + V_o^- e^{-\gamma l} \\ I_s(l) &= \frac{V_o^+}{Z_o} e^{+\gamma l} - \frac{V_o^-}{Z_o} e^{-\gamma l} \end{aligned} \right] \text{ So } Z(l) = \frac{V_s(l)}{I_s(l)}$$

$$\left[Z(l) = \left(\frac{V_o^+ e^{+\gamma l} + V_o^- e^{-\gamma l}}{V_o^+ e^{+\gamma l} - V_o^- e^{-\gamma l}} \right) Z_o \right]$$

Impedance on any point of transmission line



$$Z(l) = \frac{V_S(l)}{I_S(l)}$$

$$\left[Z(l) = \frac{V_0^+ e^{\gamma l} + V_0^- e^{-\gamma l}}{V_0^+ e^{\gamma l} - V_0^- e^{-\gamma l}} Z_0 \right] \Rightarrow Z(l) = Z_0 \frac{V_0^+ e^{\gamma l} \left[1 + \frac{V_0^-}{V_0^+} e^{-2\gamma l} \right]}{V_0^+ e^{\gamma l} \left[1 - \frac{V_0^-}{V_0^+} e^{-2\gamma l} \right]}$$

$$S. \quad Z(l) = Z_0 \frac{[1 + \Gamma(l)]}{[1 - \Gamma(l)]}$$

$$Z(l) = Z_0 \frac{[1 + \Gamma(l)]}{[1 - \Gamma(l)]}$$

$$\Gamma(l) = \frac{Z(l) - Z_0}{Z(l) + Z_0}$$

Imp. at load $l=0$

$$\Gamma(0) = \frac{Z_L - Z_0}{Z_L + Z_0}$$

Characteristics of transmission line



$$Z(l) = Z_0 \left[\frac{1 + \Gamma(l)}{1 - \Gamma(l)} \right] \quad \cdot \quad \Gamma(l) = \frac{V_0^-}{V_0^+} e^{-2\gamma l}$$

$$\Gamma(l) = \frac{Z(l) - Z_0}{Z(l) + Z_0}$$

$$\Gamma_{l(l=0)} = \frac{V_0^-}{V_0^+}$$

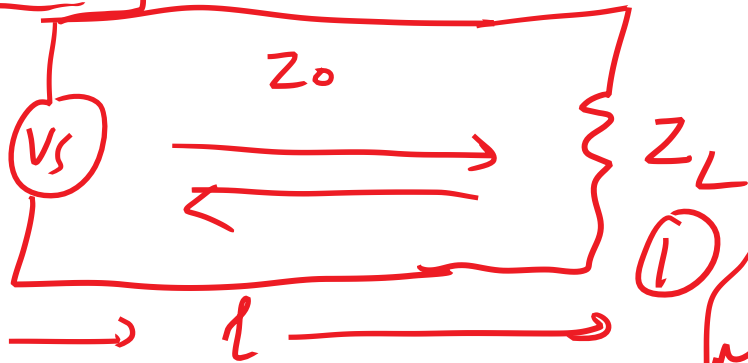
$$\Gamma_{l \text{ at } l=0} = \frac{Z_L - Z_0}{Z_L + Z_0} \quad \text{at } l=0 \quad Z_L = \frac{Z_0(1 + \Gamma_L)}{(1 - \Gamma_L)}$$

$$0 \leq |\Gamma| \leq 1$$

Characteristics of transmission line



Cases:



$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} \quad Z_L = Z_0$$

$$\Gamma_L = 0$$

No reflection

matched line

$$\begin{aligned} \Gamma(l) &= \Gamma_L e^{-2\gamma l} \\ &= \Gamma_L e^{-2\alpha l - j\beta l} \end{aligned}$$

② Infinite line

There will be no reflection